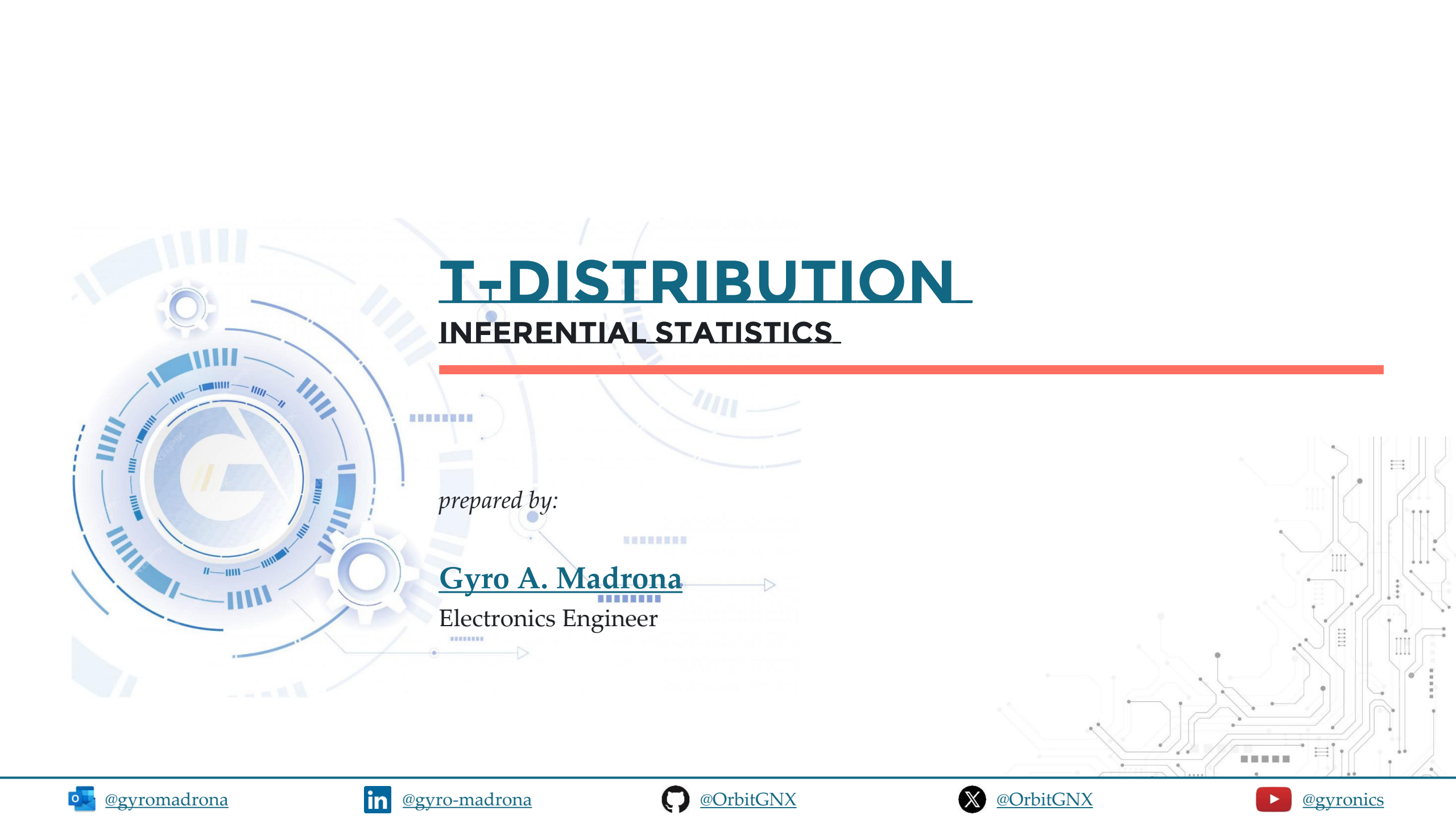




T-DISTRIBUTION

INFERENTIAL STATISTICS

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TOPIC OUTLINE

t-Distribution



T-DISTRIBUTION



T-DISTRIBUTION

The t-distribution (Student's t-Distribution) is used to calculate the confidence interval when:

1. The population variance (σ^2) is unknown.
2. The sample size is small ($n < 30$).

Formula

$$CI = \bar{x} \pm t_{n-1, \alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

where

$\bar{x}$ = sample mean

$t_{n-1, \alpha/2}$ = t-statistic

$\frac{s}{\sqrt{n}}$ = standard error

t-table

d.f. / α	0.1	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
inf.	1.282	1.645	1.960	2.326	2.576
CI*	80%	90%	95%	98%	99%

z-statistic



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syntax

Percent point function **t.ppf()** returns the critical **t** value for a given probability (1-alpha).

```
from scipy import stats
```

Two-Tailed Test

```
t_critical = stats.t.ppf(1-alpha/2, dof)
```

One-Tailed Test

```
t_critical = stats.t.ppf(1-alpha, dof)
```



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syntax

Interval function t.interval() returns a tuple (lower_limit, upper_limit) representing the confidence interval for t-interval.

```
from scipy import stats

ci_lower, ci_upper = stats.t.interval(

    confidence = confidence_level,

    df = degrees of freedom (n-1),

    loc = sample_mean,

    scale = standard_error

)
```



EXERCISE

A power company measures the voltage output (in volts) of a batch of transformers. A random sample of 10 transformers is tested, and their voltage outputs are recorded in "transformer-voltage-data-10-sample" dataset. Calculate a **95% confidence interval** for the true mean voltage output (μ) of all transformers.

Solution

Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{X} = \frac{120.2 + 94.4 + \dots + 139}{10}$$

$$\bar{X} = 119.44$$



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$$\bar{x} = 119.44$$

Solution

Sample standard deviation

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$s^2 = \frac{(120.2 - 119.44)^2 + \dots + (139 - 119.44)^2}{10-1}$$

$$s^2 = 112.62$$

$$\underline{s = 10.61}$$



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$$\bar{x} = 119.44$$

$$s = 10.61$$

Solution

t-statistic

$$\text{dof} = \cancel{n-1}^{10}$$

$$\text{dof} = 9$$

$$\alpha = 0.05 \text{ for } 95\% \text{ confidence level}$$

$$t_{n-1, \alpha/2}$$

$$\underline{t_{9, 0.025} = 2.262} \text{ critical value}$$

↗
from t-table



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$$\bar{x} = 119.44$$

$$s = 10.61$$

$$t_{9, 0.025} = 2.262$$

Solution

Confidence interval

$$CI = \bar{x} \pm t_{n-1, \alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

$$CI = 119.44 \pm 2.262 \left(\frac{10.61}{\sqrt{10}} \right)$$

$$CI = [111.85, 127.03]$$

ans

95% confidence that the true mean voltage output is between 111.85V and 127.03V



EXERCISE

You are an inventory manager at a retail store that sells shoes. The company wants to optimize its inventory levels to ensure that each shoe size is always in stock while minimizing overstocking. To achieve this, you need to determine the **95% confidence interval** for the monthly demand of each shoe size based on historical sales data from the past year.

dataset

shoe-inventory-dataset.csv

Solution



LABORATORY

